SIT1001 - Week 3 Unit 2 Part 1: Discrete and continuous random variables

2.1 Random Variables of the Discrete Type

Revision:

Example 2.1 (Ex 3.1-1 H&G) A rat is selected at random from a cage and its sex is determined. The set of possible outcomes is female and male. Thus, the outcome space is

$$S = \{\text{female, male}\} = \{\text{F, M}\}.$$

Let X be a function defined on S such that

$$X = 0 \text{ for female}$$

$$X = 1 \text{ for male}$$

Thus, X is a real-valued function that has the outcome space S as its domain and the set of real numbers $\{x; x = 0, 1\}$ as its range.

We call X **a random variable** and, in this example, the **space associated with X** is $\{x; x = 0, 1\}$

> ***Definition 2.1 (Def 3.1-1 H&G)*** *Given a random experiment with an outcome space S, a function X that assigns to each element s in S one and only one real number X(s) = s is called* ***random variables****. The* ***space of X*** *is the set of real numbers {x; X(s) = x, s ∈ S}, where s∈S means the element s belongs to the set S.*

Example 2.2 (Ex 3.1-2 H&G) Let the random experiment be the cast of a die, observing the number of spots on the side facing up.

(a) Determine the outcome space associated with this experiment
(b) For each s ∈S, let $X(s) = s$. State the space of the random variable X.
(c) For each s ∈S, let $Y(s) = s^2$. State the space of the random variable Y.

Solution:

There are two major difficulties in example above:

1. In many practical situations the probabilities assigned to the events are unknown.
2. Since there are many ways of defining a function X on S, which function do we want to use?

Definition 2.2 (Def 3.1-2 H&G) *The* ***probability mass function*** *(p.m.f) f(x) of a discrete random variable X is a function that satisfies the following properties:*

(a) $f(x) > 0, \quad x \in S;$

(b) $\sum_{x \in S} f(x) = 1;$

(c) $P(X \in A) = \sum_{x \in A} f(x),$ *where* $A \subset S$

Example 2.3 (Ex 3.1.3 H&G) Roll a four-sided die twice and let X equal the larger of the two outcomes if they are different and the common value if they are the same.

(a) Determine the outcome space associated with this experiment

(b) For each s $\in S$, let $X(s) = s$. State the space of the random variable X.

(c) Find $P(X = x)$ for every x in the space of X.

(d) Hence, state the pmf of X.

Solution:

2.2 Type of variable

Two distinct types of variable:

a. **Discrete random variable**

It is a random variable that assumes countable values. For example,

1. The number of enrolment per academic year
2. The number of syringe missing per box.
3. The number of patients died of dengue per month.

b. **Continuous random variable**

It is a random variable that can assume any value contained in one or more intervals. For example,

1. The weight of a person
2. The marks in an examination
3. The daily composite index

3.2.1 Probability distribution of a discrete random variable

The **probability distribution** of a discrete random variable lists all the possible values that the random variable can assume and their probabilities. The probability distribution can be presented in the form of a table, graph or mathematical formula.

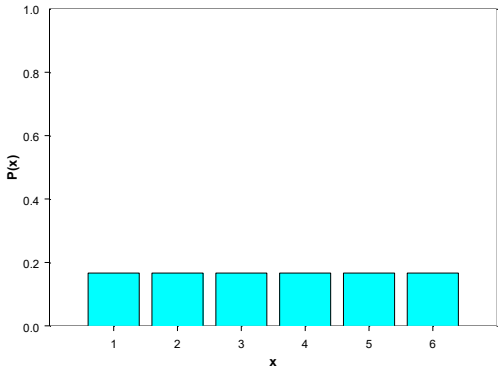
For example, If you throw a dice once, variable *x* may take value 1, 2, 3, 4, 5 or 6.

(a) Tabulation form

x	1	2	3	4	5	6	
P(*X*=*x*)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\sum P(x) = \frac{6}{6} = 1$

(b) Graphical form

(c) mathematical formulation



EMBED Equation.3

Example 2.4 *(Probability problem)*

If you throw a dice once, variable *x* may take value 1, 2, 3, 4, 5 or 6.

x	1	2	3	4	5	6	
P(*X*=*x*)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\sum P(x) = \frac{6}{6} = 1$

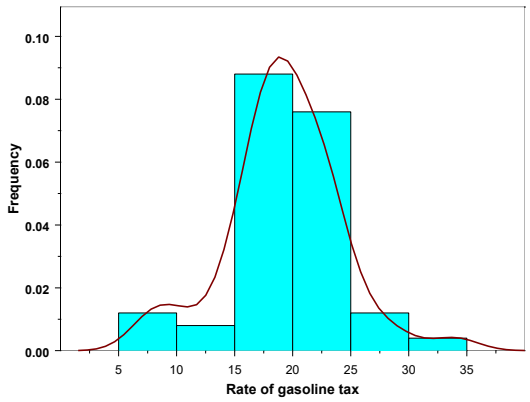
Find

1. P(X = 2) =
2. $P(X \geq 2)$ =
3. $P(2 \leq X < 5)$ =

3.2.2 Probability distribution of a continuous random variable

A continuous random variable is specified by its **probability density function (pdf)** written as $f(x)$.

One way to display continuous r.v. is using histogram. Then, a smoothed polygon is used as an approximation of the probability distribution curve of the r.v. This curve is also called the probability density function of the r.v.



If X is a continuous r.v. with pdf $f(x)$ valid over the range $a \le x \le b$, then

1. $\int_{\text{all } x} \boldsymbol{f(x)}\, dx = 1$. That is, $\int_a^b \boldsymbol{f(x)}\, dx = 1$, which means the area under the curve $y = f(x)$ is 1.

2. If $a \le x_1 \le x_2 \le b$, then $P(x_1 \le X \le x_2)$ = $\int_{x_1}^{x_2} \boldsymbol{f(x)}\, dx$.

Example 2.5 (Probability problem)

A continuous r.v. has a pdf $f(x)$ where $f(x) = \frac{x}{8}$, $0 \le x \le 4$. Find

a. P(X = 2) =

b. $P(X \ge 2)$ =

c. $P(2 \le X < 5)$ =

2.3 Cumulative distribution

Cumulative distribution sums or accumulates the probabilities found by means of the density.

Definition 2.3 (Def 3.2.2 Milton &Arnold). Let X be a **discrete/continuous** r.v. with density f. The cumulative distribution function for X, denoted by F, is defined as

$$F(x) = P(X \le x) \quad \text{for } x \text{ real}$$

Special case for continuous variable: Obtaining *f* from *F* in the continuous case

$$F'(x) = f(x)$$

Example 2.6 (Exer 7, chap 3, M&A)

The density for discrete variable X is given in Table 2.1.

X	0	1	2	3	4	5
$f(x)$	0.7	0.2	0.05	0.03	0.01	0.01

(a) Find $f(5)$.
(b) Find the table for F.
(c) Use F to find $P(1 \le X \le 3)$.
(d) Use F to verify that $P(X = 3) = 0.03$.

Example 2.7 (Ex 4.1.2 M&A)

The density for variable X is

$$f(x) = 12.5x - 1.25, \quad 0.1 \le x \le 0.5$$

(a) Find $F(x)$.
(b) Hence, find $P(0.2 \le X \le 0.3)$.

Example 2.8 (Ex 3.1.2, 3.2.1 and 3.2.2 Milton & Arnold)

(a) The basic premise underlying the field of immunology is that an animal is immunized by injection of a suitable antigen. In one study, malignant plasmacytoma cells are exposed to lymphocytes carrying a specific antigen. It is hoped that these cells will fuse, because the fused cells retain the ability to grow continuously and also to retain the antibody characteristics of the antigen fused. In this way, the animal is quickly immunized. Cells are exposed to the lymphocytes one at a time in the presence of polyethylene glycol, a fusion-promoting agent. Let Y denote the number of cells exposed to obtain the first fusion.

(i) Is Y random?

(ii) If yes, state the space of Y.

(b) We kow that under the condition described in (a), the probability that a given cell will fuse is 0.5. It is also reasonable to assume that the cells behave independently.

(i) Find the probability the first cell will fuse.

(ii) Find the probability that the first cell will not fused but the second one will fuse, i.e., P(Y=2). Similarly, find P(Y=3) and P(Y=4).

(iii) Hence or otherwise, state the probability mass function of Y.

(iv) What is the probability that we will need to expose four or more cells to antigen-carrying lymphocytes in the presence of polyethylene glycol to obtain the first fusion?

Example 2.9 (Ex 4.1.1 Milton & Arnold)

Consider the graph of $f(x)$ as given in Figure 1.

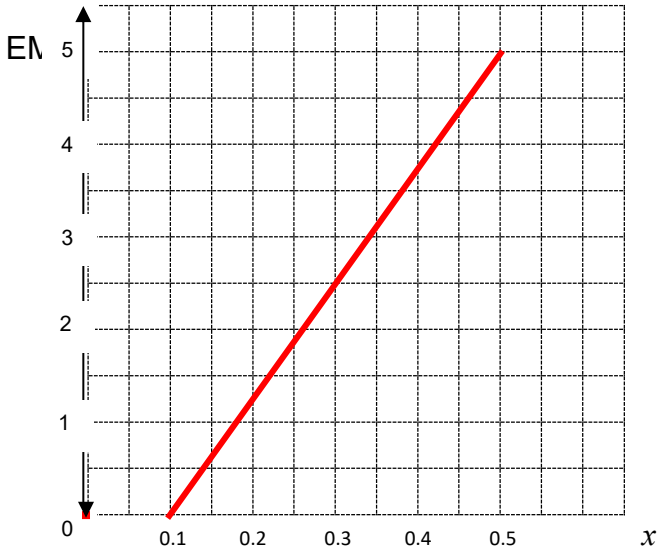


Figure 2.1

The lead concentration in gasoline currently ranges from 0.1 to 0.5 grams per litter. Let *X* denote the number of grams of lead per litter of gasoline and the density of X is given in Figure 2.1

(a) State the density of *X* in mathematical form.

(b) What is the probability that the lead concentration in a randomly selected litter of gasoline will lie between 0.2 and 0.3 grams inclusive?